Do US macro releases move Bitcoin?

A high-frequency causal analysis of CPI, NFP and FOMC announcements, 2022–2026

×2.4

volatility in the 30 min after a release

−65 bps

per σ of surprise, 5 min after

p = 0.002

parallel trends rejected

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github.com/lschwob/causal-inference-proj

Roadmap

1. **Question** — Is it the release, or just a busy hour? — *Léonard*
2. **Data** — 151 releases, 1-minute Bitcoin prices, public surprises — *Léonard*
3. **Design** — DAG, naive vs DiD, parallel trends, synthetic day — *Julius*
4. **Results** — Volatility, direction, pre-release drift — *Alexandro*
5. **Limits** — Which assumptions hold, which do not — *Alexandro*
6. **Decision** — What a desk should do with this — *All*

Is it the release, or just a busy hour?

Bitcoin moves a lot around **08:30 New York** (CPI, NFP) and **14:00** (FOMC). Is that **caused** by the releases, or would it move anyway at that hour, in that market?

- **Unit:** a 4-hour window around one release (−120 to +120 min, 5-min grid)
- **Treatment:** a scheduled release at t0 · **Outcome:** BTC path in bps and its volatility
- **Estimand: ATT.** ATE and ATU are not identified: no day without a release ever gets one

Fundamental problem

On a release day we see Bitcoin *with* the release, never *without* it.

Everything that follows is about **building that missing counterfactual** credibly.

Takeaway. We estimate the effect of a release **on release days** (ATT), for volatility and for direction.

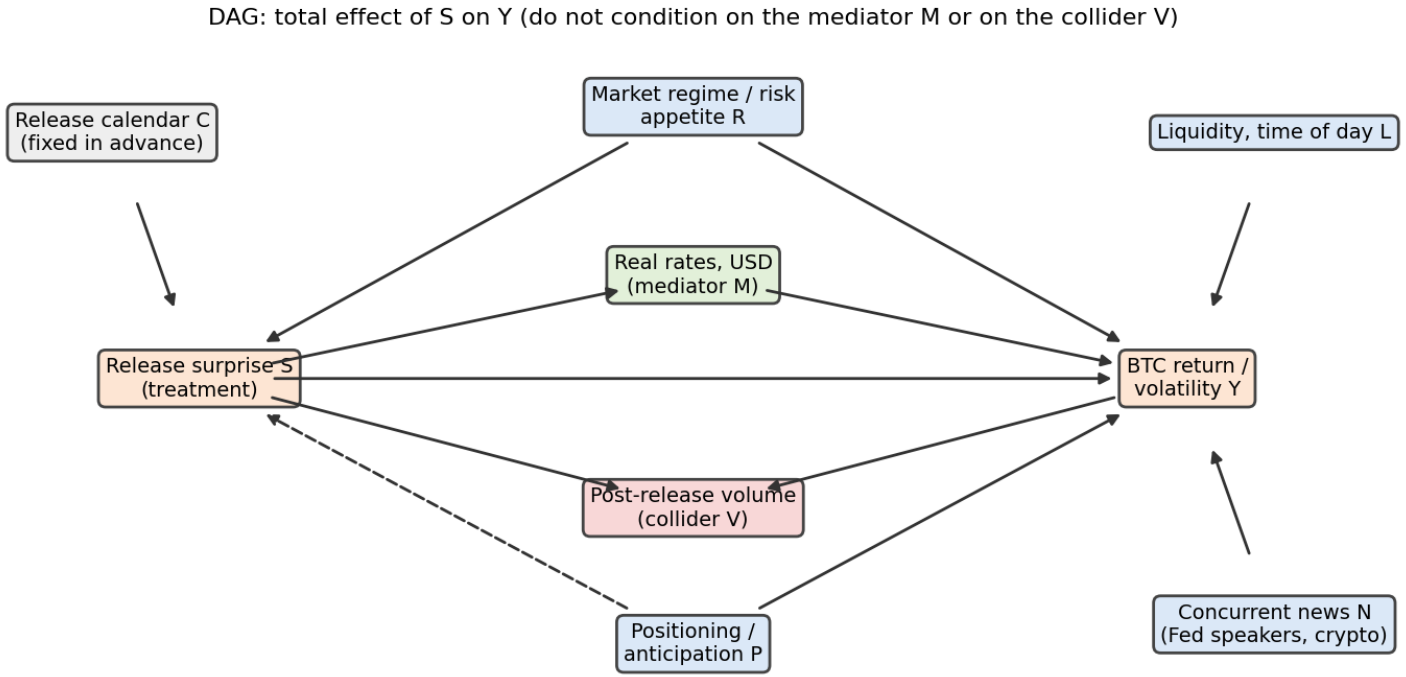
Public data: 151 releases, 2022–2026

Source	What	Coverage
Binance	BTCUSDT 1-minute prices	2.5 M bars, Jan 2022 – Oct 2026
FRED, Federal Reserve	Release calendar	56 CPI · 57 NFP · 38 FOMC
Investing.com	Consensus → surprise = actual − consensus	NFP 57/57 · core CPI 55/56
FRBSF (Bauer–Swanson)	FOMC policy surprise	16/38 decisions (2022–23)

- Surprises are standardised **within each type** (1 σ of that release): we compare signs across types, not levels
- No Bloomberg: consensus rebuilt from public sources; CPI = **core** CPI

Takeaway. Release dates are set months in advance, so their **timing** is plausibly exogenous to Bitcoin.

What we control, and what we must not



- **Confounder R** (market regime) → compare with **donor days within ±180 days** + pre-release covariates
- **Mediator M** (real rates, USD) → **not** controlled: we want the total effect
- **Collider V** (post-release volume) → **never** conditioned on
- **Positioning P, news N** → unobserved: main threats

Takeaway. Local donor days close the regime backdoor; positioning and news cannot be closed, so we **test** them.

A naive comparison is biased

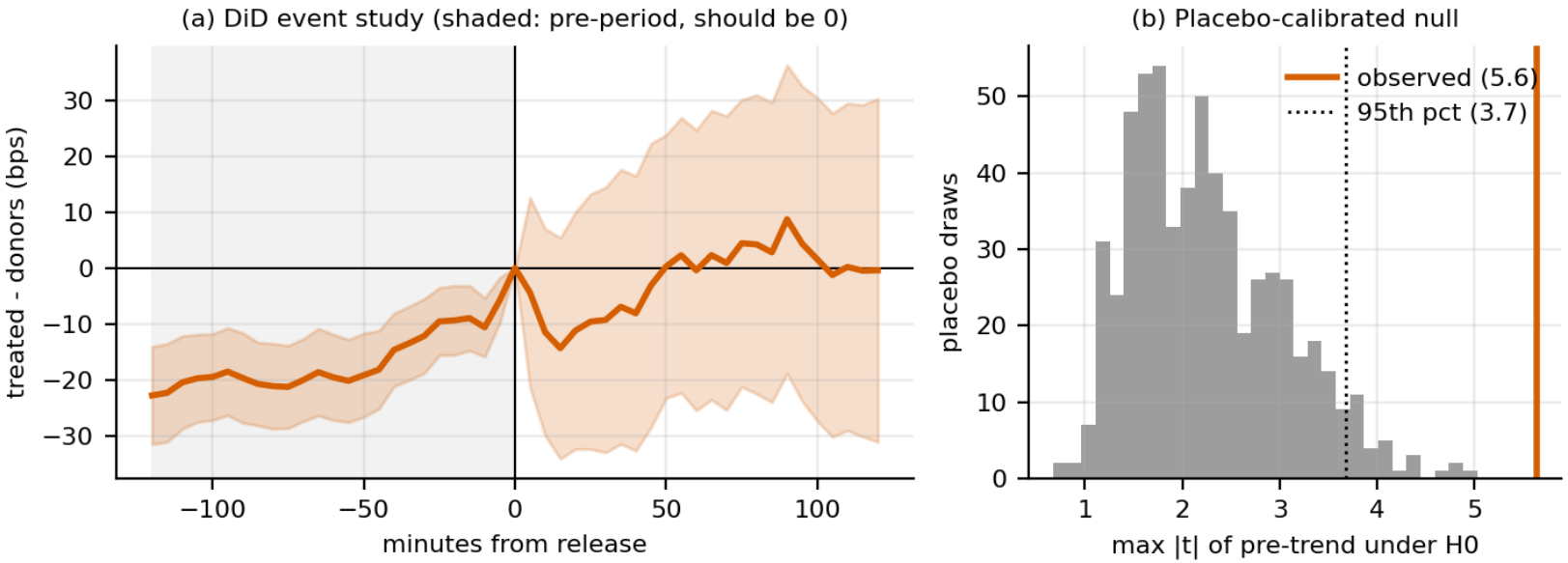
Control group = donor days: business days with no release at the same clock time, within ±180 days. Same 2×2 logic as the tutoring example from class:

Log realised volatility (30 min)	Release days before → after	Donor days before → after	Naive “after only”	DiD
08:30 · CPI, NFP (113)	3.38 → 4.34	3.30 → 3.29	1.05	**0.97**
14:00 · FOMC (38)	3.70 → 4.31	3.42 → 3.42	0.89	**0.62**

- Volatility multiplier = exp(DiD): **×2.6** at 08:30, **×1.9** at 14:00. The matched DiD (donors within ±180 days of each release) gives ×2.4 overall
- “After only” mixes the effect with the gap that already existed before t0 (selection bias); “before vs after” would fail if volatility rose at that hour anyway, which donor days show it does not

Takeaway. FOMC days are **already more volatile before 14:00**: “after only” wrongly credits that to the decision. The DiD removes it.

Parallel trends are rejected

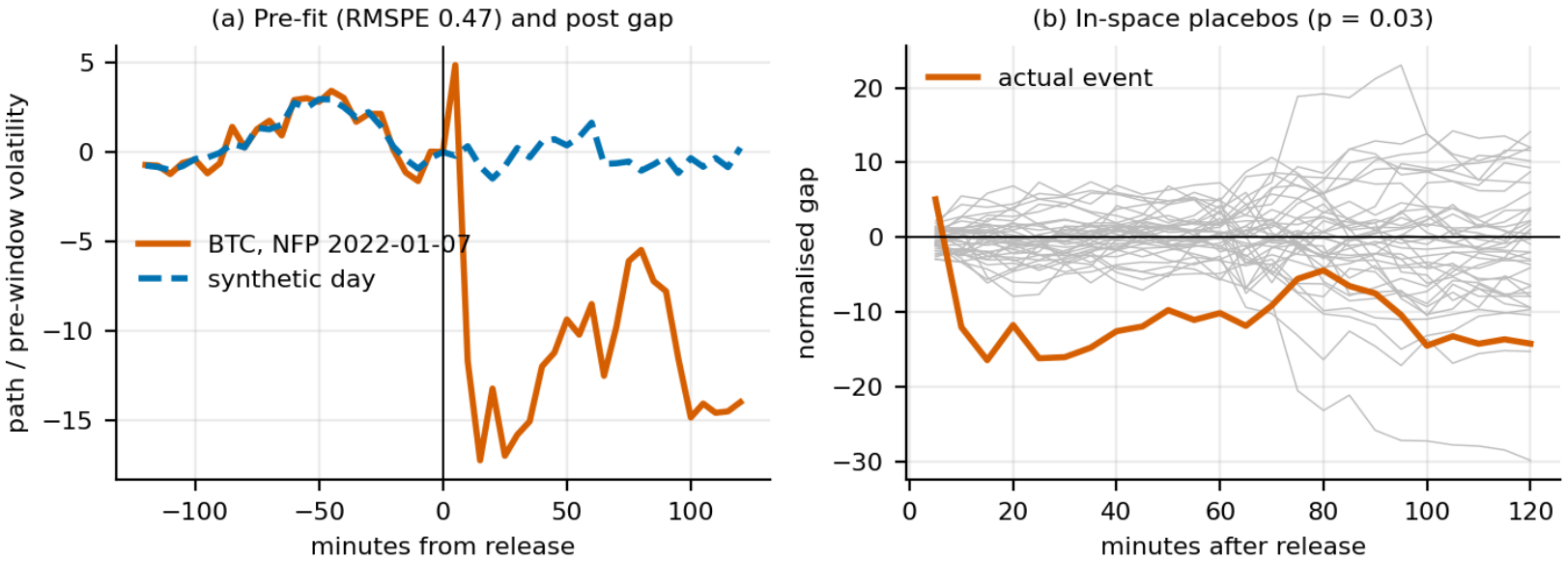


- Before t0 the gap should be zero; BTC rises ≈ **23 bps** in the 2 hours before
- Joint test max|t| = 5.6, placebo-calibrated **p = 0.002** (08:30: 0.003 · 14:00: 0.010)
- DiD still fine for **volatility**

Left: treated − donors gap, relative to t0 (grey = before: should be 0). Right: placebo null of the pre-trend test.

Takeaway. Bitcoin drifts up **before** the release, so the equal-weight DiD is not credible for signed effects.

Main method: the “synthetic day”



- A **weighted mix of donor days** that tracks BTC before t0 (ridge; penalty tuned on placebo days only)
- **In-space placebos:** every donor day plays a fake treated unit
- Classical SC (weights ≥ 0, sum 1) agrees: correlation 0.99

Left: actual BTC and its synthetic day (NFP, 7 Jan 2022). Right: actual gap vs 40 placebo gaps.

Takeaway. Each release gets its own counterfactual, and its own placebo-based p-value.

Result 1: releases multiply volatility

30-min realised volatility

×2.4

95% CI ×2.2–×2.7 · t = 17 · 151 releases

Significant vs placebos

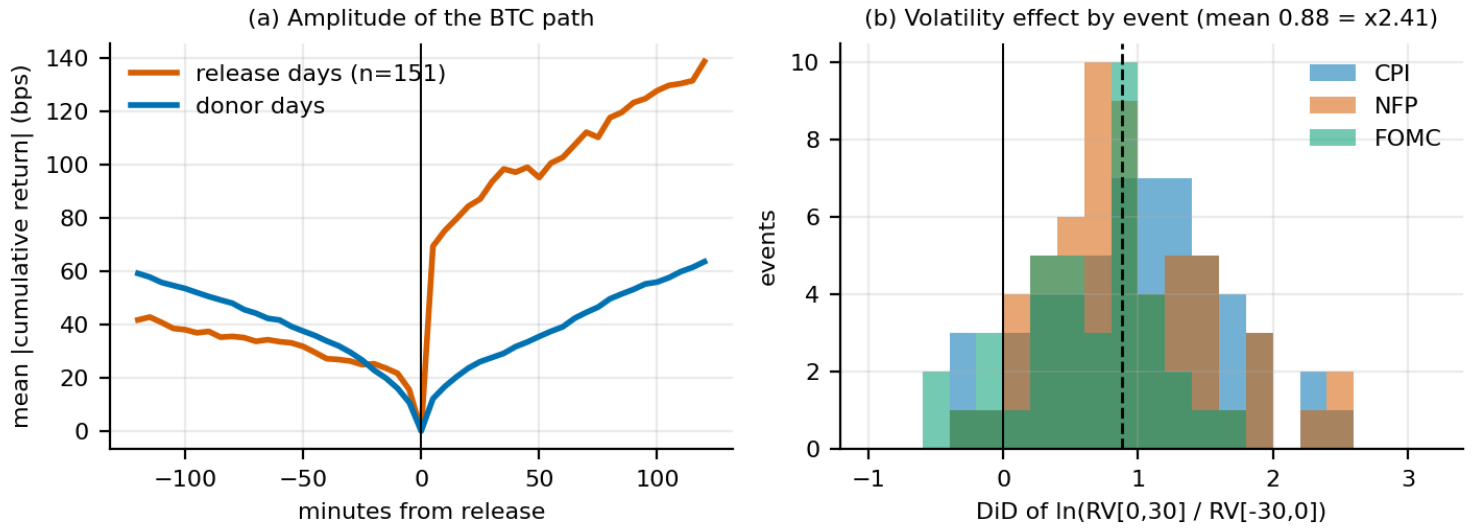
41–44%

of releases, vs 5% expected by chance

Clock shifted by ±6–12 h

≈ 0

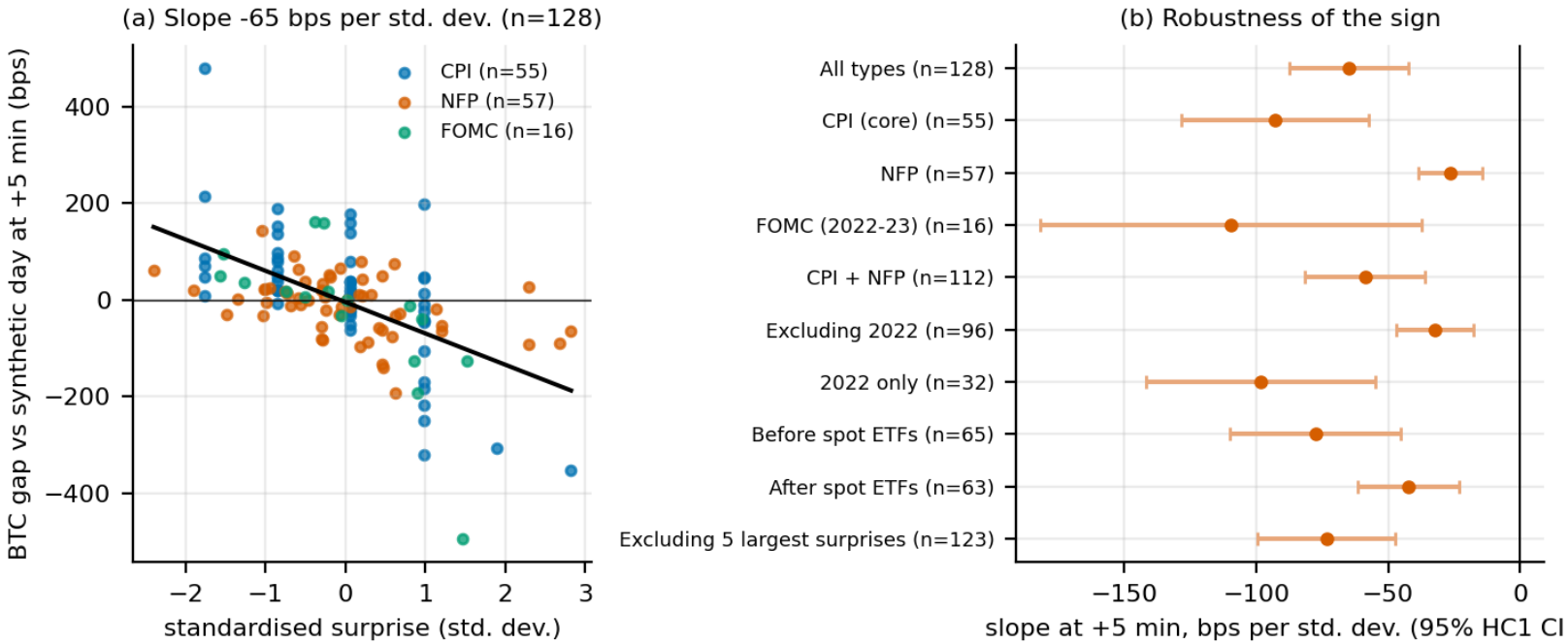
0.04 · 0.02 · −0.03 vs 0.97 at the true clock



- By type: CPI ×2.7 · NFP ×2.5 · FOMC ×1.9
- Not driven by a few extreme events

Takeaway. The effect is large, shows up release by release, and **disappears when the clock is shifted**.

Result 2: the reaction follows the surprise

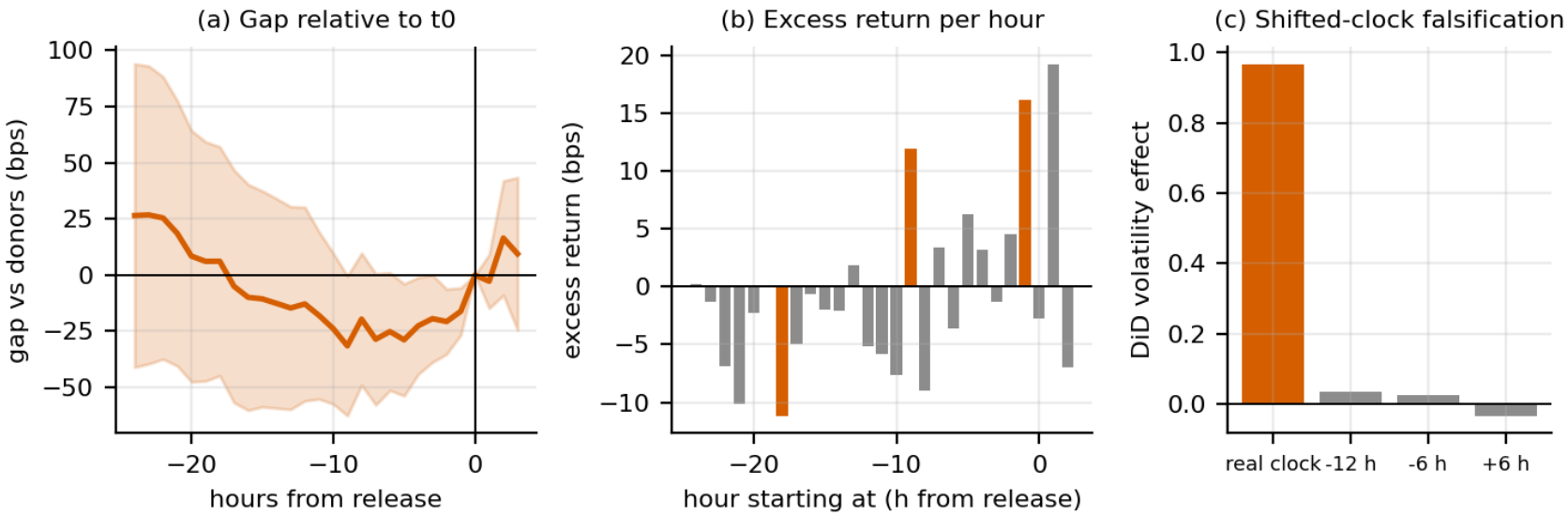


- **−65 bps per σ** at 5 min, −80 at 30 min (128 releases, permutation $p < 0.001$)
- Negative in **20/20** sub-samples; robust to outliers (Theil–Sen −49)
- **Not stable:** −98 in 2022, −32 since

Left: surprise vs BTC gap at +5 min. Right: slope by sub-sample (95% CI).

Takeaway. A **more hawkish** print than expected makes Bitcoin **fall**. The sign is robust; the size is not.

Before the release: a drift we cannot explain



- Concentrated in the **last hour**: +16 bps (t = 3.0)
- Only at the true clock; robust to the donor pool (±30 to ±365 days)
- Positioning? Leaks? Calendar effect?

08:30 releases. (a) gap vs donors by hour · (b) excess return per hour · (c) volatility effect, true vs shifted clocks.

Takeaway. Real and specific to the release time, but its **cause is not demonstrated**: we report it, we do not trade it.

What could make us wrong

Assumption	Verdict	Evidence
Exogenous calendar	**Plausible**	Dates set in advance; the content depends on the economy
Parallel trends	**Rejected**	Pre-release drift → synthetic day as main method
No anticipation	**Doubtful**	Drift in the last hour; cause unknown
No interference (SUTVA)	**Partial**	Releases share donors → placebo-calibrated SEs
No concurrent news	**Untestable**	Fed speakers, crypto news, ETF flows
Surprise measurement	**Imperfect**	Core CPI only, FOMC 16/38, public consensus
Stable regime	**No**	Effect weaker after 2022 and after spot ETFs

Takeaway. Each assumption is tied to a test in the notebook. The volatility result survives them; direction is more fragile.

Recommendation: manage the risk, not the signal

DO

- Cut size or leverage before 08:30 New York (14:00 on FOMC days)
- Widen stops and risk limits for the first 30 minutes
- Avoid executing large orders in that window

DON'T (YET)

- Trade direction systematically: size unstable, needs out-of-sample tests and costs
- Trade the pre-release drift: cause unknown
- Above-consensus print → hedge longs, rather than open shorts

Takeaway. Yes for risk management, not yet for a directional strategy.

Four things to remember

1. Releases **cause** Bitcoin volatility: ×2.4 in 30 minutes, falsification passed.
2. Direction follows the surprise (hawkish → BTC down): sign robust, size unstable.
3. Parallel trends fail: the synthetic day is the credible counterfactual.
4. Use it for risk management, not (yet) as a trading signal.

Questions?

Code, notebook (runs on Colab), report: github.com/lschwob/causal-inference-proj